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THE ABSORPTION LINES WITHIN THE HYDROGEN EMISSION OF MIRA CETI

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ABSTRACT

On spectrograms taken with the high dispersion of the coudé spectrograph, most of the absorption minima within the emission lines of Mira Ceti may be identified with metallic lines of the reversing layer of the star. Sixty-five lines were thus assigned to thirteen elements, and four other lines in $H\gamma$ were attributed to *TiO*. The measurements seem to furnish sufficient proof that the hydrogen producing the emission lines is at a lower level than the source of the numerous metallic absorption lines of the star's atmosphere.

In 1899, W. W. Campbell[1] reported that the bright hydrogen lines $H\gamma$ and $H\delta$ in the three-prism spectra of *o* Ceti were not single lines but were made up of three components. W. H. Wright[2] found but two components of $H\gamma$ at the following maximum, but in 1909 the three components were again seen. Three components of both lines were recorded[3] in 1925 on Mount Wilson spectrograms taken with the three-prism spectrograph. On coudé spectrograms of high dispersion, taken before or shortly after maximum light, similar abnormalities have been seen in each of the hydrogen lines observed in Mira Ceti and other long-period Me variables. The broad emission lines seem to be broken up into several components by absorption lines having different widths and intensities. The appearance of some of the lines is well shown by reproductions of coudé spectrograms[4] and by the microphotometer curves of Figures 1 and 2. An obvious and perhaps sufficient explanation of the peculiar structure of the hydrogen lines is that they originate at a comparatively low level in the star's atmosphere and are so greatly widened that the overlying atoms of other elements of the reversing layer absorb narrow portions of the outgoing radiation in the characteristic lines of those elements. The apparent components of the bright lines are then the remaining wave lengths of emission not affected by such absorption lines as happen to occur at positions intermediate between the extremities of the hydrogen radiation. This explanation was, in principle, advanced many years ago by Miss A. Clerke[5] to account for the weakness and disappearance of $H\epsilon$ in the spectrum of *o* Ceti at certain phases and was later applied by C. D. Shane[6] to the weakened ultraviolet hydrogen lines $H\zeta$, $H\kappa$, $H\lambda$, $H\mu$, and $H\xi$. P. W. Merrill[7] strongly supports this theory as a general explanation of the mutilation of the emission lines in long-period variables. He was able to identify some of the absorption minima within $H\zeta$, $H\eta$, $H\theta$, and $H\iota$ with lines which might be expected in the reversing layer; he also suggested that

[1] *Ap. J.*, **9**, 31, 1899.

[2] *Lick Obs. Bull.*, **6**, 60, 1910.

[3] A. H. Joy, *Mt. W. Contr.*, No. 311; *Ap. J.*, **63**, 303, 1926.

[4] P. W. Merrill, *Spectra of Long-Period Variable Stars* (Chicago: University of Chicago Press, 1940), opp. p. 57 (*H* 11, U Ori; $H\delta$ and $H\gamma$, *o* Cet); *Mt. W. Contr.*, No. 713; *Ap. J.*, **102**, opp. p. 349, 1945 ($H\gamma$ and $H\delta$, U Ori); *Pub. A.S.P.*, **57**, opp. p. 179, 1945 (*H* 10 and *H* 11, *o* Cet); W. S. Adams, *Mt. W. Contr.*, No. 638, *Ap. J.*, **93**, opp. p. 14, 1941 (*H* 16, *o* Cet).

[5] *Problems in Astrophysics* (London, 1903), p. 226.

[6] *Lick Obs. Bull.*, **10**, 132, 1922.

[7] *Spectra of Long-Period Variable Stars*, p. 72; *Pub. A.S.P.*, **57**, 178, 1945.

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1947CMWCI.737....1J

the low intensity of $H\alpha$, $H\beta$, and $H\gamma$ might be due to the absorption of the *TiO* bands. W. S. Adams[8] identified certain lines of iron, chromium, and vanadium with the absorption lines in $H\delta$, $H\zeta$, H 10, and H 16.

An extensive series of spectrograms of Mira Ceti obtained by Dr. P. W. Merrill and the writer during the years from 1934 to 1947 now makes it possible to study this interesting problem in greater detail than has hitherto been possible. Eighty-eight spectrograms were used in measuring the wave lengths of the absorption lines within the hydrogen emission lines. Of these, 75 were taken with the coudé spectrograph of the 100-inch telescope and 13 with the three-prism ultraviolet spectrograph at the 60-inch. The approximate mean dispersions are: 40 plates, 3 A/mm; 33 plates, 10 A/mm; 10 plates, 15 A/mm; and 5 plates, 20 A/mm. Nineteen of the spectrograms extend to H 18, λ 3696, in the ultraviolet, but only a few scattered measures were possible in the complicated region of shorter wave length, where the distinction between bright lines and spaces separating absorption lines is difficult.

The means of the measured wave lengths of the absorption lines within and near the emission lines of hydrogen are shown in Table 1. For each spectrogram the measured values were corrected for the radial velocity of the star as determined from the other absorption lines of the same plate or of other plates taken at about the same date. Only such lines as occur within about 1 A of the center of the hydrogen lines are included in the table. Since the total width of the strong hydrogen emission is usually not much greater than 1 A, some of the absorption lines may fall on the edge of, or partly outside, the principal emission.

The intensities in the third column of Table 1 are rough relative estimates made from spectrograms where the lines are easily seen. The estimates are greatly influenced by the density of the near-by emission. For such lines as are evidently blends the other contributing line is in the last column. Variations in the intensity of the absorption lines during the cycle of the light-variation are largely masked by the changes in the position and strength of the emission lines. The microphotometer curves of Figures 1 and 2 illustrate the appearance of the absorption lines within or near some of the hydrogen lines at phases before and after maximum. The curves for $H\gamma$ are at four phases of the same cycle.

The identifications in the fifth and sixth columns are based chiefly on agreement in wave length, the average difference being 0.036 A, or about 3 km/sec in terms of velocity. This difference includes the error in determining the velocity of the star and the error in the laboratory wave length, as well as the errors of measurement of the particular line in the star. The intensities in the seventh column are from the *M.I.T. Wavelength Tables.* Although only about half the 65 identifications pertain to strong or well-known lines, the excellent agreement of the wave lengths points to the probability that most of the identifications are justified. The distribution of the identifications among the elements is quite reasonable for a star of low temperature. The number of lines identified for the several elements are: *Ca*, 1; *Ti* I, 8; *Ti* II, 1; *V*, 14; *Cr*, 4; *Mn*, 1; *Fe*, 24; *Co*, 3; *Ni*, 5; *Y*, 1; *Zr*, 1; *Mo*, 1; *In*, 1. The identification of the *In* line λ 4101.764, as suggested by P. W. Merrill for U Orionis[9] and R Leonis,[10] is interesting and provides a mechanism by which electrons of indium atoms may be raised to the level at which they can emit the line λ 4511.310. A search for similar fluorescent effects in other elements was not fruitful.

The 24 *Fe* lines identified within the bright hydrogen lines show the same correlation between displacement and excitation potential as that found by W. S. Adams[11] for *Fe* lines in o Ceti and by P. W. Merrill[12] in U Orionis, R Serpentis, and R Hydrae. The mean

[8] *Mt. W. Contr.*, No. 638; *Ap. J.*, **93**, 18, 1941.

[9] *Mt. W. Contr.*, No. 713; *Ap. J.*, **102**, 352, 1945.

[10] *Mt. W. Contr.*, No. 720; *Ap. J.*, **103**, 285, 1946.

[11] *Mt. W. Contr.*, No. 638; *Ap. J.*, **93**, 19, 1941.

[12] *Mt. W. Contr.*, No. 713; *Ap. J.*, **102**, 348, 1946; *Mt. W. Contr.*, No. 717; *Ap. J.*, **103**, 6, 1946.

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1947CMWCI.737....1J

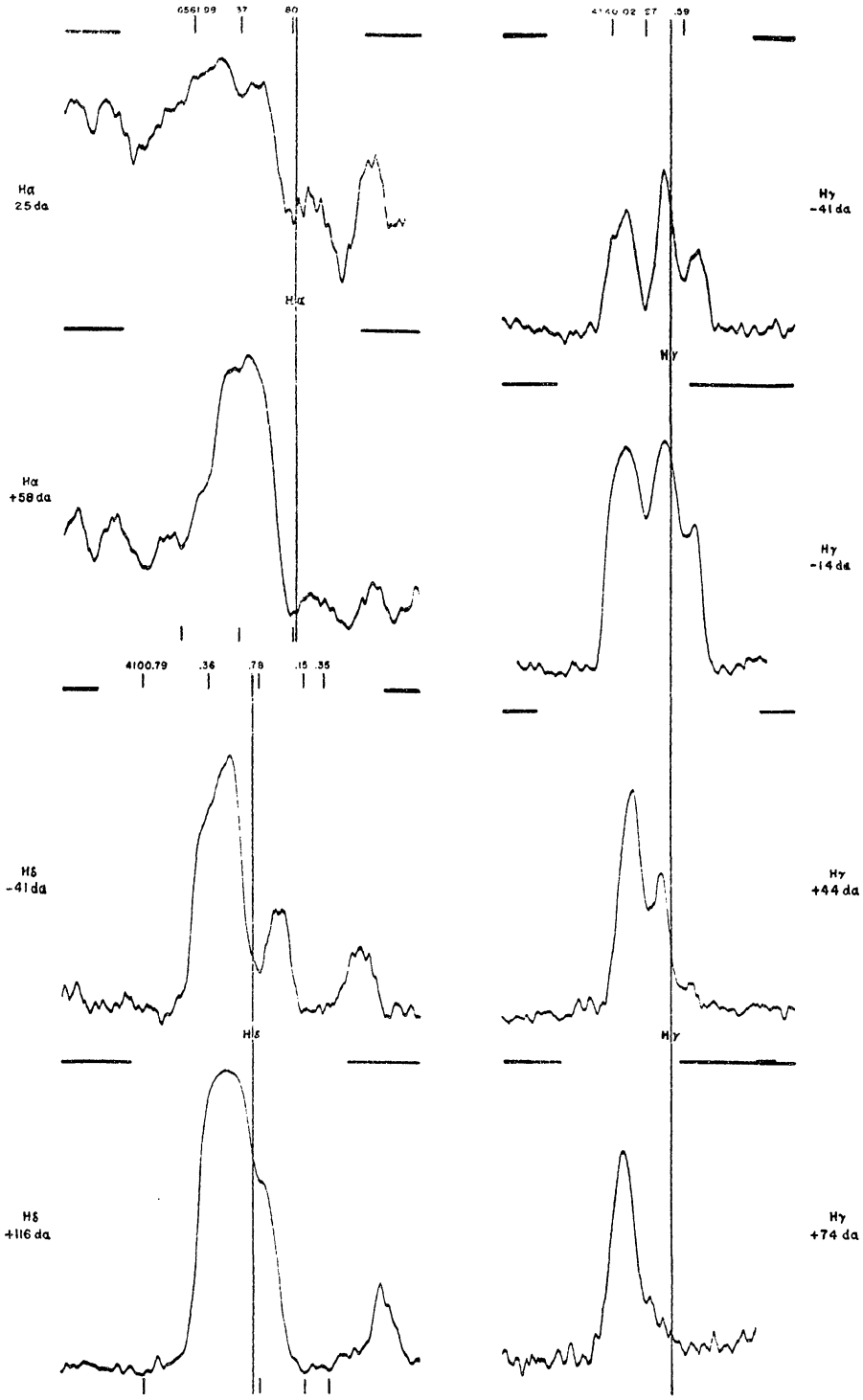

FIG. 1.—Microphotometer curves of *o* Ceti, showing the emission lines $H\alpha$ and $H\delta$ at phases before and after maximum, and $H\gamma$ at four different phases of the same cycle. The vertical line indicates the undisplaced (laboratory) position of the hydrogen line, and the numbers above are the wave lengths of the absorption lines.

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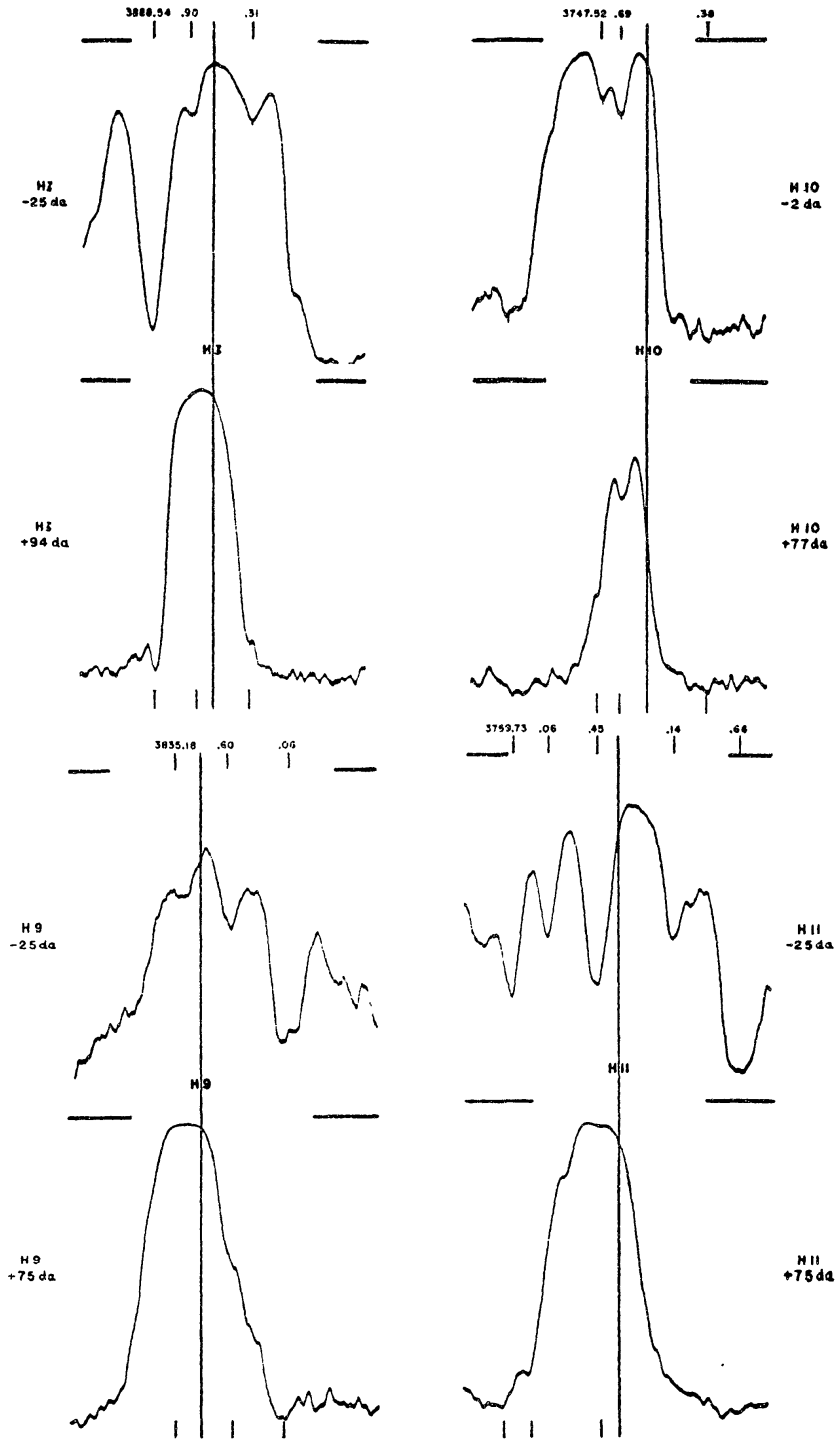

FIG. 2.—Microphotometer curves of *o* Ceti showing the emission lines $H\zeta$, $H9$, $H10$, and $H11$ at phases before and after maximum. The vertical line indicates the undisplaced (laboratory) position of the hydrogen line, and the numbers above are the wave lengths of the absorption lines.

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1947CMWCI.787....1J

TABLE 1

WAVE LENGTHS AND IDENTIFICATION OF ABSORPTION LINES WITHIN OR NEAR THE HYDROGEN EMISSION OF *o* CETI

Hydrogen Line	Mean Measured λ	Int.	No. Plates	Identification λ	Element	M.I.T. Int.	Multiplet No.	Remarks
Hα	6561.982	2	4	.....	.....	.....	.....	
	62.369	2	3	.....	.....	.....	.....	
	62.801	5	5	.....	.....	.....	.....	
Hγ	4340.022	1	3	4340.011	*TiO*	.....	.....	
	40.271	2	40	40.304	*TiO*	.....	.....	
	40.593	4	40	40.593	*TiO*	.....	.....	
	41.015	4	2	40.986	*TiO*	.....	.....	4341.013 *V* 60 (5)
Hδ	4100.792	5	5	4100.745	*Fe*	80	18	
	01.358	2	6	01.272	*Fe*	40	698	
	01.784	10	42	01.764	*In*	2000	1	Identified by P. W. Merrill
	02.153	2	3	02.159	*V*	30	41	
	02.347	3	4	02.376	*Y*	150	7	
Hζ	3887.969	2	9	3888.020	*Ti*	15	175	
	88.545	10	42	88.517	*Fe*	400	45	Identified by W. S. Adams
	88.901	1	19	88.825	*Fe*	40	488	Identified by W. S. Adams
	89.313	2	14	.....	.....	.....	.....	
H 9	3834.673	1	4	3834.735	*Cr*	25	70	
	35.183	1	5	35.185	*V*	30	.....	
	35.598	2	29	35.560	*V*	50	44	
	36.055	5	28	36.054	*V*	10	44	
H 10	3797.520	2	9	3797.517	*Fe*	300	607	Identified by W. S. Adams
	97.690	3	46	97.716	*Cr*	100	139	Identified by W. S. Adams
	98.381	4	12	98.259	*Mo*	1000	1	3798.513 *Fe* 400 (21)
H 11	3769.730	3	17	3769.736	*Ti*	10	.....	
	70.061	1	18	69.995	*Fe*	80	387	
	70.450	10	40	70.405	*Fe*	4	177	
	71.143	2	24	.....	.....	.....	.....	
	71.657	5	30	71.652	*Ti*	70	17	
H 12	3749.526	5	16	3749.487	*Fe*	1000	21	Identified by C. D. Shane
	50.306	1	2	50.349	*Cr*	20	27	
	50.786	1	9	50.763	*Mn*	60	24	
H 13	3733.336	12	22	3733.319	*Fe*	400	5	
	34.849	10	23	34.867	*Fe*	1000	21	Identified by C. D. Shane
H 14	3721.386	1	6	3721.358	*V*	10	11	3721.396 *Fe* 3 (131)
	21.638	1	10	21.632	*Ti* II	60	13	3721.606 *Fe* 6 (437)
	22.016	1	6	21.998	*V*	70	91	
	22.564	15	20	22.564	*Fe*	500	5	Identified by C. D. Shane
H 15	3711.340	4	23	3711.411	*Fe*	50	494	
	12.110	1	14	12.177	*Co*	40	84	
H 16	3703.003	2	23	3702.942	*Ti*	20	132	
	03.603	3	28	03.584	*V*	200	29	Identified by C. D. Shane
	04.107	1	10	04.060	*Co*	300	35	
H 17	3696.685	1	18	.....	.....	.....	.....	Unidentified molecular band

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1947CMWCI.737....1J

TABLE 1—*Continued*

Hydrogen Line	Mean Measured λ	Int.	No. Plates	Identification λ	Identification Element	Identification M.I.T. Int.	Identification Multiplet No.	Remarks
H 17	97.172	1	15	.....	.....	.....	.....	Unidentified molecular band
	97.657	2	15	.....	.....	.....	.....	Unidentified molecular band
H 18	3690.762	1	5	3690.730	*Fe*	80	807	
	91.136	1	11	.....	.....	.....	.....	Unidentified molecular band
	91.509	1	18	.....	.....	.....	.....	Unidentified molecular band
	91.888	1	14	.....	.....	.....	.....	Unidentified molecular band
	92.263	3	11	3692.225	*V*	200	29	
H 19	3686.040	2	1	3685.998	*Fe*	150	385	
	87.379	12	3	87.458	*Fe*	400	21	
H 20	3682.293	2	3	3682.226	*Fe*	400	772	
	83.052	5	5	83.047	*Co*	200	99	
H 21	3678.974	2	3	3678.98	*Fe*	5	124	
	79.861	20	3	79.915	*Fe*	500	5	
H 22	3675.716	3	2	3675.700	*V*	100	29	
	76.352	1	2	76.314	*Fe*	200	228	
	76.631	1	1	76.684	*V*	300	115	
	77.050	2	1	77.085	*V*	25	.....	
H 23	3674.098	12	3	3674.15	*Ni*	200	32	
H 24	3670.814	1	1	3670.810	*Fe*	12	133	
	71.230	2	1	71.205	*V*	100	70	
	71.653	5	4	71.672	*Ti*	150	19	
H 25	3668.994	3	1	3668.965	*Ti*	100	18	
	69.204	3	1	69.241	*Ni*	150	2	
	69.580	1	2	69.523	*Fe*	200	291	
	70.371	6	1	70.427	*Ni*	150	4	
H 26	3667.105	5	1	3667.252	*Fe*	80	570	3666.944 *Fe* (46)
	67.747	1	1	67.741	*V*	80	114	
	68.024	2	1	67.999	*Fe*	60	438	
	68.444	1	1	.....	.....	.....	.....	
H 27	3665.990	1	1	3665.980	*Cr*	20	48	
	66.176	3	1	66.19	*Cr*	12	46	
H 28	3664.049	10	1	3664.095	*Ni*	300	4	
	64.618	3	1	64.537	*Fe*	35	391	
	65.223	2	2	65.142	*V*	100	115	
H 30	3661.931	4	1	3661.951	*Ni*	50	16	
H 31	3661.271	5	3	3661.20	*Zr*	18	12	
H 32	3660.558	8	3	3660.631	*Ti*	90	18	
H 35	3658.043	8	3	3658.097	*Ti*	150	19	

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1947CMWO.33...1J

displacement for E.P. 0.53 volts is —0.6 km/sec, based on laboratory wave lengths, while for E.P. 2.85 volts it is +2.8 km/sec. This result indicates that the *Fe* lines within the hydrogen emission have the same characteristics as those outside.

The probable occurrence of the identified lines has also been checked by noting the presence in the star's spectrum of other lines of similar intensity belonging to the same multiplets.

Of the 81 lines measured, only 3 remain unidentified, aside from the lines of $H\alpha$, H 17, and H 18, which are probably molecular. The lines in $H\gamma$, which were first observed in 1898 by W. W. Campbell and have since been measured on a large number of spectrograms, have, until the present time, defied identification. The failure to trace the origin of these best-known features of the $H\gamma$ emission in Mira Ceti has doubtless delayed the acceptance of the hypothesis of absorption by gases of the reversing layer—a hypothesis which now seems entirely probable. The individual absorption lines of the *TiO* bands, however, seem to furnish the needed source of absorption for these lines of $H\gamma$. Two laboratory spectrograms of *TiO*, dispersion 2.7 A/mm, kindly supplied by Dr. R. B. King, were measured in the region of $H\gamma$. Four *TiO* band lines have mean wave lengths as follows: λλ 4340.011, 4340.304, 4340.593, 4340.986. These lines agree well in wave length with the 4 lines of $H\gamma$, measured in the star. The last line is probably largely due to the vanadium line, λ 4341.013. Similar identifications with *TiO* would doubtless result from measures of laboratory plates in the region of $H\alpha$ and $H\beta$. A well-marked overlying molecular band whose origin has not yet been traced is responsible for six unidentified lines in H 17 and H 18. The stellar lines λ 3696.885 *Ti* 20 (177) and λ 3697.426 *Fe* 100 (389) should be present but seem to be lost in the band structure.

Wave lengths determined at different phases of the star's light-variation show no significant correlation with phase, indicating that the changes in the emission strata have little effect on the stability of the absorbing layers. On the assumption that there is no variation of wave length with phase, the average deviation of a single plate from the mean measured wave length of a line is 0.030 A or 2.1 km/sec for the lines at $H\gamma$ and $H\delta$. The high accuracy obtained for the mean wave lengths gives considerable confidence in the identifications suggested and proves beyond question that absorption by atoms of the reversing layer sufficiently explains the abnormalities of the bright hydrogen lines in Mira Ceti and other long-period Me variables.

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